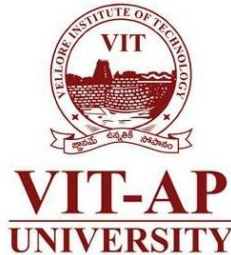


MAT1001 Calculus for Engineers

Double Integration and Polar Coordinates



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Theorem

A function f is said to be integrable on a finite closed interval $[a, b]$ if the limit

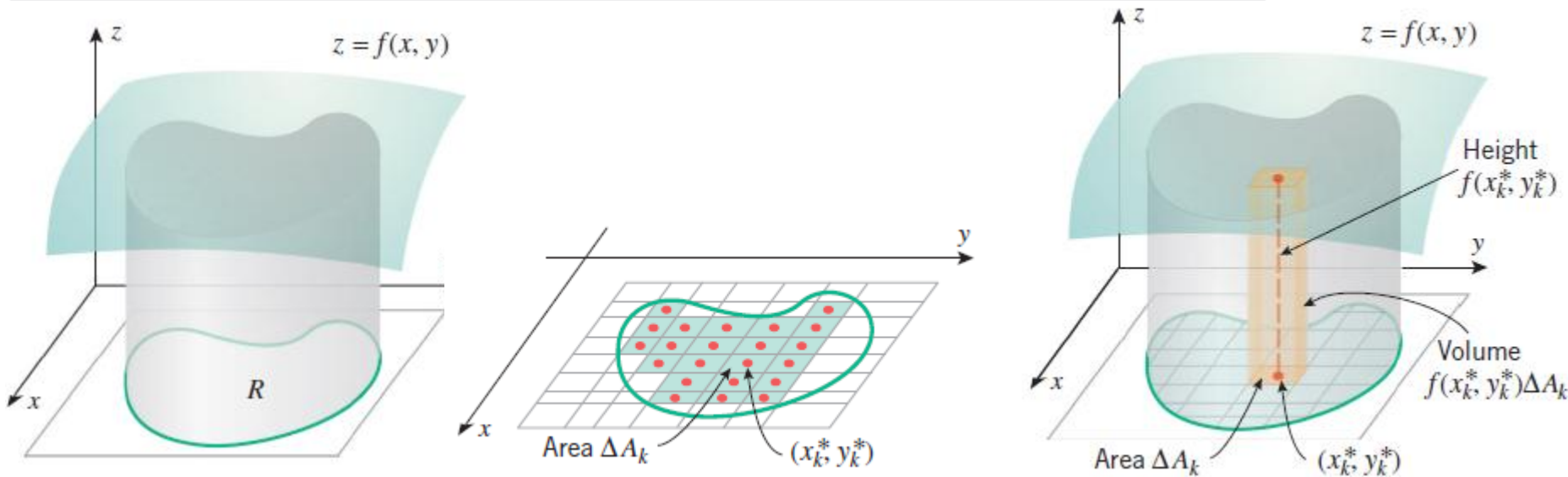
$$\lim_{\max \Delta x_k \rightarrow 0} \sum_k f(x_k^*) \Delta x_k$$

exists and independent of x_k^* . Then we write it by

$$\int_a^b f(x) dx = \lim_{\max \Delta x_k \rightarrow 0} \sum_k f(x_k^*) \Delta x_k$$

Double Integral – Volume Under a Surface

THE VOLUME PROBLEM Given a function f of two variables that is continuous and nonnegative on a region R in the xy -plane, find the volume of the solid enclosed between the surface $z = f(x, y)$ and the region R



$$V = \lim_{n \rightarrow +\infty} \sum_{k=1}^n f(x_k^*, y_k^*) \Delta A_k$$

Double Integral – Volume Under a Surface

DEFINITION (Volume Under a Surface) If f is a function of two variables that is continuous and nonnegative on a region R in the xy -plane, then the volume of the solid enclosed between the surface $z = f(x, y)$ and the region R is defined by

$$V = \lim_{n \rightarrow +\infty} \sum_{k=1}^n f(x_k^*, y_k^*) \Delta A_k \quad (2)$$

Here, $n \rightarrow +\infty$ indicates the process of increasing the number of subrectangles of the rectangle enclosing R in such a way that both the lengths and the widths of the subrectangles approach zero.

$$\iint_R f(x, y) dA = \lim_{n \rightarrow +\infty} \sum_{k=1}^n f(x_k^*, y_k^*) \Delta A_k \quad \text{is called the } \textit{double integral} \text{ of } f(x, y) \text{ over } R.$$

Evaluation of Double Integration

$$\int_c^d \int_a^b f(x, y) dx dy = \int_c^d \left[\int_a^b f(x, y) dx \right] dy$$

$$\int_a^b \int_c^d f(x, y) dy dx = \int_a^b \left[\int_c^d f(x, y) dy \right] dx$$

► **Example** Evaluate

$$\int_1^3 \int_2^4 (40 - 2xy) dy dx \quad R \text{ defined by } 1 \leq x \leq 3 \text{ and } 2 \leq y \leq 4.$$

Solution

$$\begin{aligned} \int_1^3 \int_2^4 (40 - 2xy) dy dx &= \int_1^3 \left[\int_2^4 (40 - 2xy) dy \right] dx \\ &= \int_1^3 (40y - xy^2) \Big|_{y=2}^4 dx \\ &= \int_1^3 [(160 - 16x) - (80 - 4x)] dx = \int_1^3 (80 - 12x) dx = 112 \end{aligned}$$

Double Integration on a Rectangular Region

THEOREM (Fubini's Theorem) Let R be the rectangle defined by the inequalities

$$a \leq x \leq b, \quad c \leq y \leq d$$

If $f(x, y)$ is continuous on this rectangle, then

$$\iint_R f(x, y) dA = \int_c^d \int_a^b f(x, y) dx dy = \int_a^b \int_c^d f(x, y) dy dx$$

► **Example** Evaluate

$$\int_2^4 \int_1^3 (40 - 2xy) dx dy \quad R \text{ defined by } 1 \leq x \leq 3 \text{ and } 2 \leq y \leq 4.$$

Solution

$$\begin{aligned} \int_2^4 \int_1^3 (40 - 2xy) dx dy &= \int_2^4 \left[\int_1^3 (40 - 2xy) dx \right] dy = \int_2^4 (40x - x^2y) \Big|_{x=1}^3 dy \\ &= \int_2^4 [(120 - 9y) - (40 - y)] dy = \int_2^4 (80 - 8y) dy = (80y - 4y^2) \Big|_2^4 = 112 \end{aligned}$$

Example

Consider the solid S bounded above by the surface $z = 40 - 2xy$ and below by the rectangle R defined by $1 \leq x \leq 3$ and $2 \leq y \leq 4$.

By the method of slicing the volume of S is given by

$$V = \int_1^3 A(x) dx$$

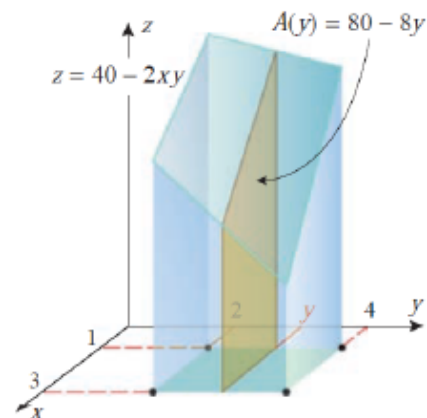
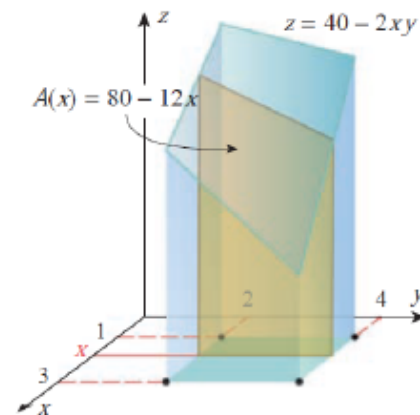
where $A(x)$ is the area of a vertical cross section of S

$$A(x) = \int_2^4 (40 - 2xy) dy$$

$$\text{Thus, } V = \int_1^3 \left[\int_2^4 (40 - 2xy) dy \right] dx = \int_1^3 \int_2^4 (40 - 2xy) dy dx$$

Similarly,

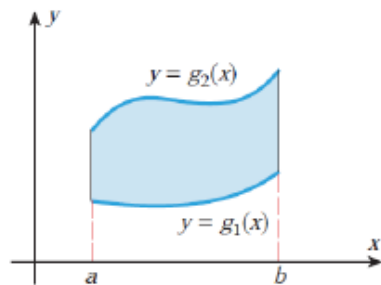
$$V = \int_2^4 A(y) dy = \int_2^4 \left[\int_1^3 (40 - 2xy) dx \right] dy = \int_2^4 \int_1^3 (40 - 2xy) dx dy$$



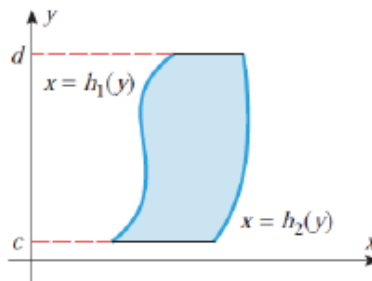
Type I and Type II Regions

DEFINITION

- (a) A **type I region** is bounded on the left and right by vertical lines $x = a$ and $x = b$ and is bounded below and above by continuous curves $y = g_1(x)$ and $y = g_2(x)$, where $g_1(x) \leq g_2(x)$ for $a \leq x \leq b$
- (b) A **type II region** is bounded below and above by horizontal lines $y = c$ and $y = d$ and is bounded on the left and right by continuous curves $x = h_1(y)$ and $x = h_2(y)$ satisfying $h_1(y) \leq h_2(y)$ for $c \leq y \leq d$



A type I region



A type II region

THEOREM

(a) If R is a type I region on which $f(x, y)$ is continuous, then

$$\iint_R f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$$

(b) If R is a type II region on which $f(x, y)$ is continuous, then

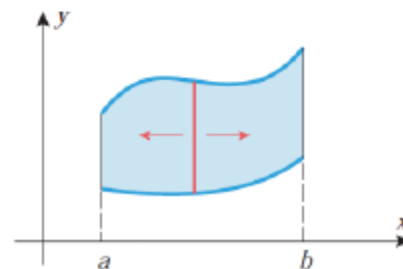
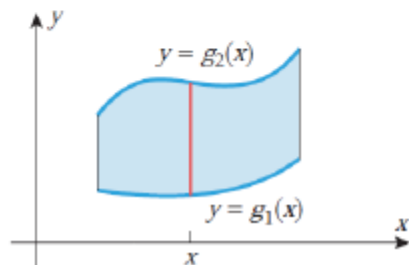
$$\iint_R f(x, y) dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy$$

Determining the Limits of Type I Region

Determining Limits of Integration: Type I Region

Step 1. Since x is held fixed for the first integration, we draw a vertical line through the region R at an arbitrary fixed value x . This line crosses the boundary of R twice. The lower point of intersection is on the curve $y = g_1(x)$ and the higher point is on the curve $y = g_2(x)$. These two intersections determine the lower and upper y -limits of integration in Formula (3).

Step 2. Imagine moving the line drawn in Step 1 first to the left and then to the right. The leftmost position where the line intersects the region R is $x = a$, and the rightmost position where the line intersects the region R is $x = b$. This yields the limits for the x -integration in Formula (3).

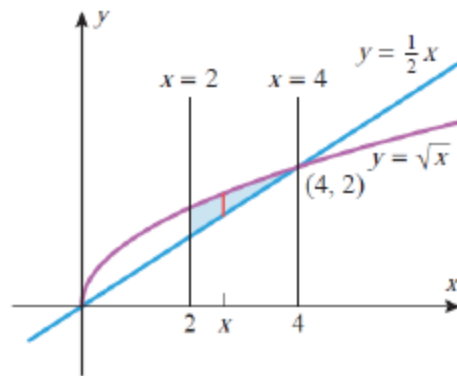


► **Example** Evaluate

$$\iint_R xy \, dA$$

over the region R enclosed between $y = \frac{1}{2}x$, $y = \sqrt{x}$, $x = 2$, and $x = 4$.

Solution. We view R as a type I region.



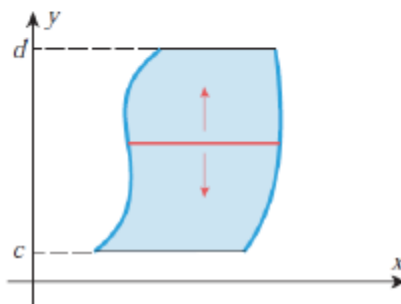
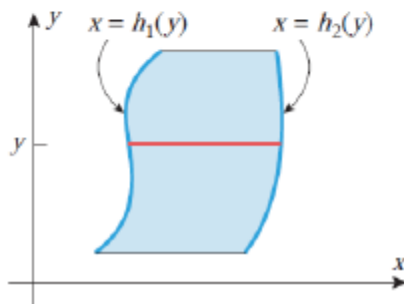
$$\begin{aligned} \iint_R xy \, dA &= \int_2^4 \int_{x/2}^{\sqrt{x}} xy \, dy \, dx = \int_2^4 \left[\frac{xy^2}{2} \right]_{y=x/2}^{\sqrt{x}} dx = \int_2^4 \left(\frac{x^2}{2} - \frac{x^3}{8} \right) dx \\ &= \left[\frac{x^3}{6} - \frac{x^4}{32} \right]_2^4 = \left(\frac{64}{6} - \frac{256}{32} \right) - \left(\frac{8}{6} - \frac{16}{32} \right) = \frac{11}{6} \end{aligned}$$

Determining The Limits of Type II Region

Determining Limits of Integration: Type II Region

Step 1. Since y is held fixed for the first integration, we draw a horizontal line through the region R at a fixed value y . This line crosses the boundary of R twice. The leftmost point of intersection is on the curve $x = h_1(y)$ and the rightmost point is on the curve $x = h_2(y)$. These intersections determine the x -limits of integration in (4).

Step 2. Imagine moving the line drawn in Step 1 first down and then up. The lowest position where the line intersects the region R is $y = c$, and the highest position where the line intersects the region R is $y = d$. This yields the y -limits of integration in (4).



Example

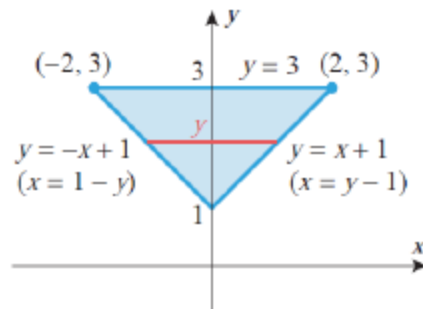
► **Example** Evaluate

$$\iint_R (2x - y^2) dA$$

over the triangular region R enclosed between the lines $y = -x + 1$, $y = x + 1$, and $y = 3$.

Solution. We view R as a type II region.

$$\begin{aligned} \iint_R (2x - y^2) dA &= \int_1^3 \int_{1-y}^{y-1} (2x - y^2) dx dy = \int_1^3 [x^2 - y^2 x]_{x=1-y}^{y-1} dy \\ &= \int_1^3 [(1 - 2y + 2y^2 - y^3) - (1 - 2y + y^3)] dy \\ &= \int_1^3 (2y^2 - 2y^3) dy = \left[\frac{2y^3}{3} - \frac{y^4}{2} \right]_1^3 = -\frac{68}{3} \quad \blacktriangleleft \end{aligned}$$



Example

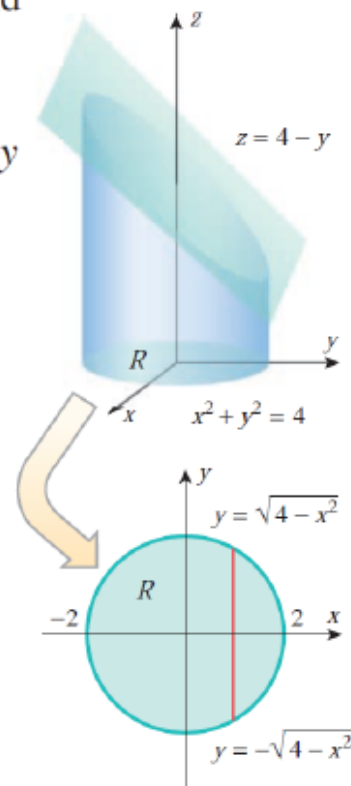
► **Example** Find the volume of the solid bounded by the cylinder $x^2 + y^2 = 4$ and the planes $y + z = 4$ and $z = 0$.

Solution. The solid shown in Figure is bounded above by the plane $z = 4 - y$ and below by the region R within the circle $x^2 + y^2 = 4$. The volume is given by

$$V = \iint_R (4 - y) dA$$

Treating R as a type I region we obtain

$$\begin{aligned} V &= \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (4 - y) dy dx = \int_{-2}^2 \left[4y - \frac{1}{2}y^2 \right]_{y=-\sqrt{4-x^2}}^{\sqrt{4-x^2}} dx \\ &= \int_{-2}^2 8\sqrt{4-x^2} dx = 8(2\pi) = 16\pi \end{aligned}$$



Reversing The Order of Integration

Sometimes the evaluation of an iterated integral can be simplified by reversing the order of integration.

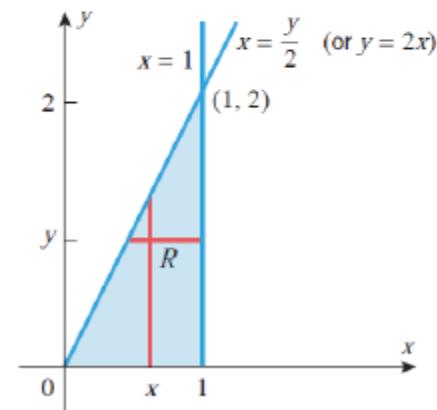
► **Example** Since there is no elementary antiderivative of e^{x^2} , the integral

$$\int_0^2 \int_{y/2}^1 e^{x^2} dx dy$$

cannot be evaluated by performing the x -integration first. Evaluate this integral by expressing it as an equivalent iterated integral with the order of integration reversed.

Solution. To reverse the order of integration, we treat R as a type I region, which enables us to write the given integral as

$$\begin{aligned} \int_0^2 \int_{y/2}^1 e^{x^2} dx dy &= \iint_R e^{x^2} dA = \int_0^1 \int_0^{2x} e^{x^2} dy dx = \int_0^1 [e^{x^2} y]_{y=0}^{2x} dx \\ &= \int_0^1 2x e^{x^2} dx = e^{x^2} \Big|_0^1 = e - 1 \end{aligned}$$



Area Using Double Integration -Example

► **Example** Use a double integral to find the area of the region R enclosed between the parabola $y = \frac{1}{2}x^2$ and the line $y = 2x$.

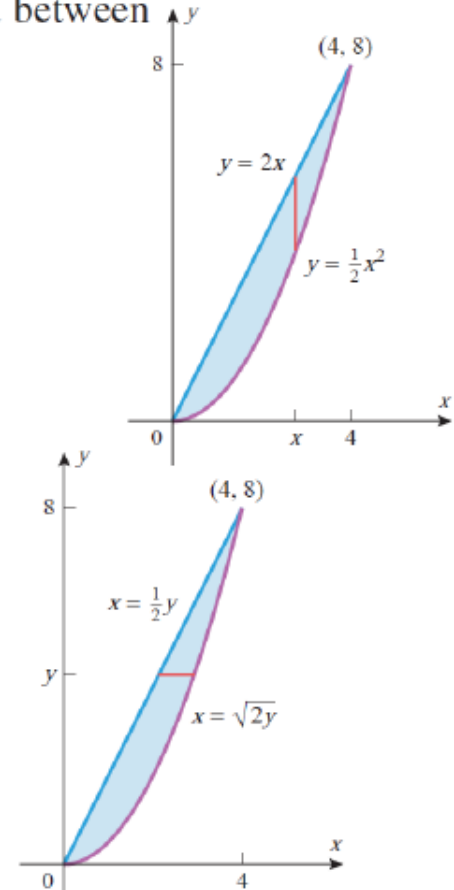
Solution.

Treating R as type I yields

$$\begin{aligned} \text{area of } R &= \iint_R dA = \int_0^4 \int_{x^2/2}^{2x} dy \, dx = \int_0^4 [y]_{y=x^2/2}^{2x} dx \\ &= \int_0^4 \left(2x - \frac{1}{2}x^2 \right) dx = \left[x^2 - \frac{x^3}{6} \right]_0^4 = \frac{16}{3} \end{aligned}$$

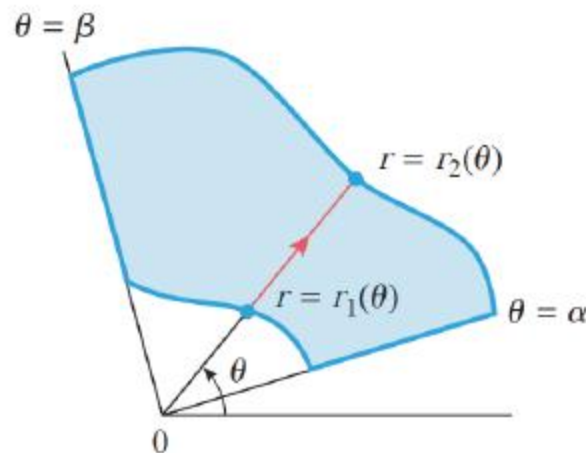
Treating R as type II yields

$$\begin{aligned} \text{area of } R &= \iint_R dA = \int_0^8 \int_{y/2}^{\sqrt{2y}} dx \, dy = \int_0^8 [x]_{x=y/2}^{\sqrt{2y}} dy \\ &= \int_0^8 \left(\sqrt{2y} - \frac{1}{2}y \right) dy = \left[\frac{2\sqrt{2}}{3}y^{3/2} - \frac{y^2}{4} \right]_0^8 = \frac{16}{3} \end{aligned}$$



THEOREM If R is a simple polar region whose boundaries are the rays $\theta = \alpha$ and $\theta = \beta$ and the curves $r = r_1(\theta)$ and $r = r_2(\theta)$ shown in Figure and if $f(r, \theta)$ is continuous on R , then

$$\iint_R f(r, \theta) dA = \int_{\alpha}^{\beta} \int_{r_1(\theta)}^{r_2(\theta)} f(r, \theta) r dr d\theta$$



Example

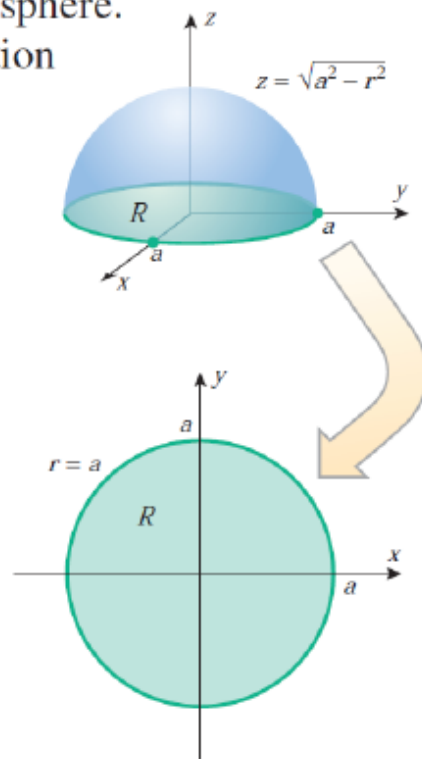
► **Example** The sphere of radius a centered at the origin is expressed in rectangular coordinates as $x^2 + y^2 + z^2 = a^2$, and hence its equation in cylindrical coordinates is $r^2 + z^2 = a^2$. Use this equation and a polar double integral to find the volume of the sphere.

Solution. In cylindrical coordinates the upper hemisphere is given by the equation

$$z = \sqrt{a^2 - r^2}$$

so the volume enclosed by the entire sphere is $V = 2 \iint_R \sqrt{a^2 - r^2} dA$
where R is the circular region. Thus,

$$\begin{aligned} V &= 2 \iint_R \sqrt{a^2 - r^2} dA = \int_0^{2\pi} \int_0^a \sqrt{a^2 - r^2} (2r) dr d\theta \\ &= \int_0^{2\pi} \left[-\frac{2}{3} (a^2 - r^2)^{3/2} \right]_{r=0}^a d\theta = \int_0^{2\pi} \frac{2}{3} a^3 d\theta \\ &= \left[\frac{2}{3} a^3 \theta \right]_0^{2\pi} = \frac{4}{3} \pi a^3 \end{aligned}$$



Converting from Rectangular to Polar Coordinates

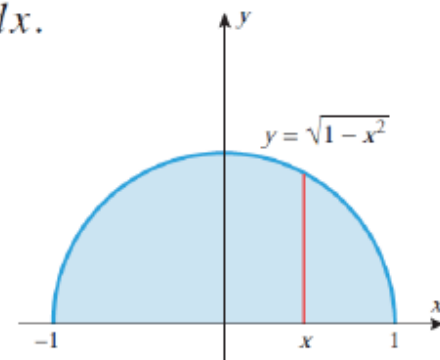
Sometimes a double integral that is difficult to evaluate in rectangular coordinates can be evaluated more easily in polar coordinates by making the substitution $x = r \cos \theta$, $y = r \sin \theta$ and expressing the region of integration in polar form; that is, we rewrite the double integral in rectangular coordinates as

$$\iint_R f(x, y) dA = \iint_R f(r \cos \theta, r \sin \theta) dA = \iint_{\text{appropriate limits}} f(r \cos \theta, r \sin \theta) r dr d\theta \quad (9)$$

► **Example 4** Use polar coordinates to evaluate $\int_{-1}^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2)^{3/2} dy dx$.

Solution.

$$\begin{aligned} \int_{-1}^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2)^{3/2} dy dx &= \iint_R (x^2 + y^2)^{3/2} dA \\ &= \int_0^\pi \int_0^1 (r^3) r dr d\theta = \int_0^\pi \frac{1}{5} d\theta = \frac{\pi}{5} \end{aligned}$$



The seal of VIT-AP University is a circular emblem. It features a central image of a temple gopuram (tower) with a flag on top, set against a background of a body of water and a city skyline. The text "VIT" is prominently displayed in the center of the emblem. Below the emblem is a banner with text in Telugu script.

Thank You...

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